

Nathan So

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EDUCATION

UC Santa Barbara – B.S. Computer Science, Physics Minor (3.94) September 2025 - June 2029

- Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Linear Algebra, Calculus
- Clubs/Events: Gaucho Racing, UCSB Aeronautics, UCSB International Capture the Flag, SB Hacks, Datathon

EXPERIENCE

Undergraduate Research Assistant – UCSB Computer Science April 2026 – Present

- Utilizing web scraping techniques to collect and structure student feedback data for school-wide research.
- Developed the data pipeline using web scraping and chrome extension scripting to facilitate research on student-instructor interaction and feedback loops in high-enrollment CS courses.

Gaucho Racing (FSAE) Firmware Member September 2025 – February 2026

- Contributed embedded C firmware for MVP CAN bus on STM32 boards for the Formula SAE electric racecar.
- Conducted hardware in the loop testing/debugging for a custom, patterned 6-foot LED under-glow using SPI.

UC San Diego - Health Hospital Volunteer May 2024 – December 2024

- Assisted a team of 6 nurses to provide care to 30+ patients in the top floor Progressive Care Unit.
- Managed patient logistics including laboratory specimen transport, discharge paperwork, and mobility assistance.

PROJECTS

Inner Circle – LA Hacks Submission April 24-26, 2026

- Architected an **automation suite** using URL payloads to programmatically generate person-specific personalized Google Calendar events and Gmail drafts, streamlining social workflow automation.
- Developed a passive-notification system and React interface to visualize social network health using Firebase.

Hey Host! – SB Hacks XII Grand Prize Winner January 10-11, 2026

- Engineered a voice-native podcast assistant using a **Next.js** frontend and **Flask** backend to enable real-time, interactive Q&A with custom AI-modeled podcaster voices.
- Built an audio pipeline integrating **faster-whisper** (STT), **porcupine** (keyword spotting), and **ElevenLabs** (TTS) for sub 3-second average response times and 85% voice recognition consistency.
- Implemented the 6 voice commands for hands-free experience, cleaned the audio transcripts with web scraping, refined the RAG pipeline to reduce API costs, and refined multiple front-end UI elements manually.

GEAR Interactive - Personal GOLD Chrome Extension September 2025 - Present

- Developed a **Chrome Extension** using **TypeScript** and **React** that injects color-coded course availability and collapsible metadata into the UCSB GOLD registration portal with strict guardrails against cross-site scripting.
- Implemented persistent state using the **Chrome Storage API** to save course history across sessions.
- Wrote an automation script to filter 12,270 course listings by GE area, time range, and seat availability, significantly reducing course information retrieval time and package size (only 2.69 MiB).

SBC Housing Explorer - UCSB Datathon February 21-22, 2026

- Architected a full-stack housing discovery platform using **React** and **Tailwind CSS** to streamline and visualize the Santa Barbara County housing search for 18 cities and Census Designated Places (CDPs).
- Developed an interactive geospatial interface with **Leaflet.js** to visualize rental properties cost burden, owning/buying availability, house-building potential, and specific occupations/demographics for each zone.
- Consisted of 8 distinct choropleth maps, 1000+ rows of housing data found online, 2000+ rows of occupation data provided by a sponsor, and a context-aware RAG chatbot that could alter the maps in real time.

SKILLS

Coding Languages: C/C++, Python, TypeScript, R, HTML, CSS, Assembly

Frameworks: React, Next.js, Flask, Selenium, Vite, Beautiful Soup, R Studio, Leaflet.js, **ASP.NET (exposure)**

Tools: Git, AI/LLM Coding Assistants, STM32, Chrome API (Storage/Tabs/Runtime), Linux environments

Practical: Food Handler's card, Hospital safety, Tutoring/Teaching, Photography, Wood Working

Interests: Beach Volleyball, , Competitive Programming, Hiking, Tennis, Weightlifting, Rock Climbing, Traveling